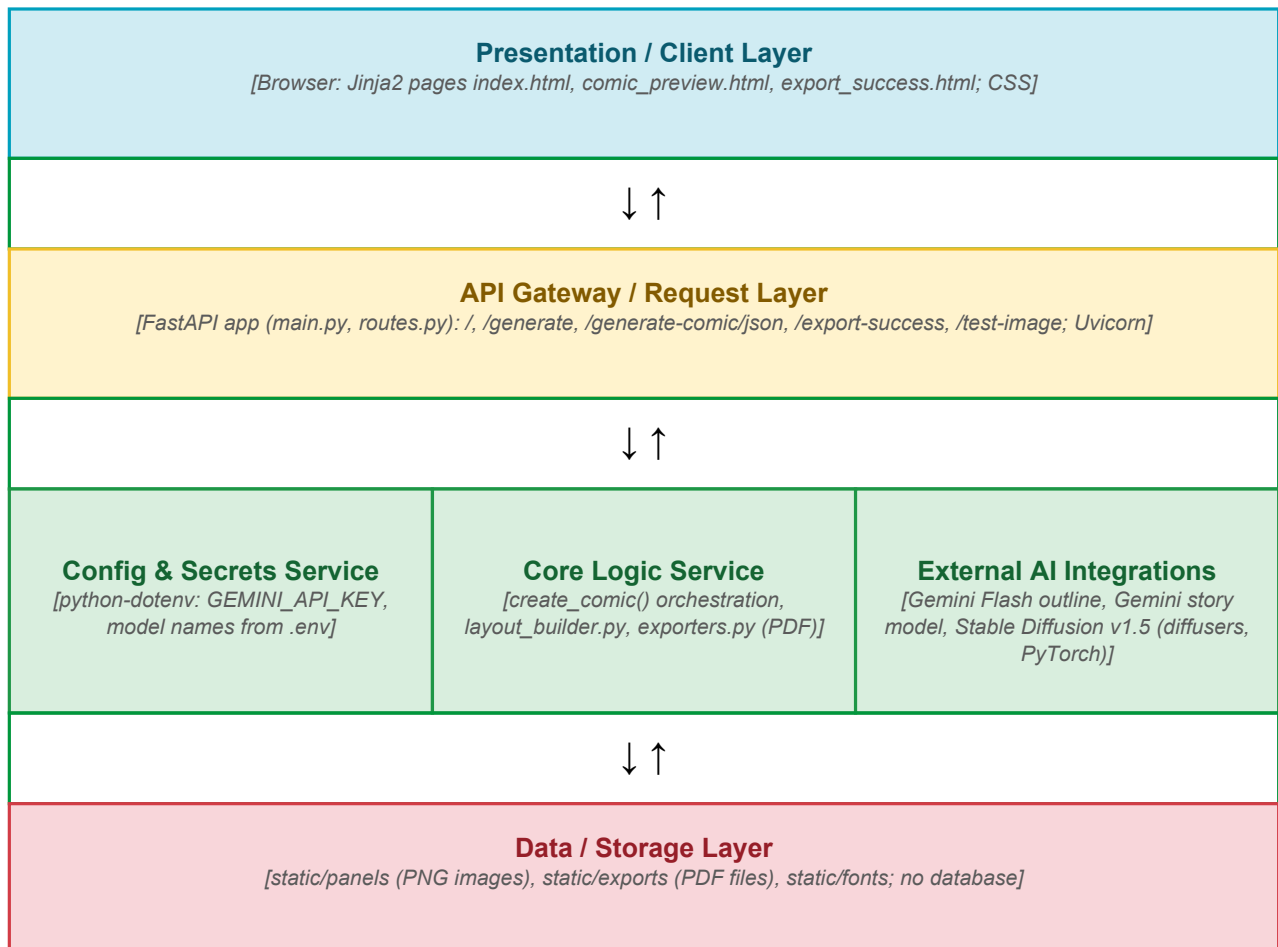


Solution Architecture

Date	2 October 2026
Team ID	12
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table:

Component Name	Description / Role	Technologies Used
Web UI	Home form, comic preview and export confirmation pages	Jinja2, HTML, CSS

FastAPI app	Mounts /static, defines routes, starts model warm-up thread	FastAPI, Uvicorn
Outline service	Returns a 5-panel JSON outline (panel, title, scene, image_prompt)	gemini_flash.py, Gemini Flash
Story service	Writes caption and narration for each panel	gemini_pro.py, Gemini
Image service	Cached Stable Diffusion pipeline; saves PNG; returns its URL	diffusers, PyTorch (MPS/CUDA/CPU)
Layout builder	Parses the story with regex and merges it with outline and images	Python (pure function)
Exporter	Writes a PDF with one page per panel	fpdf2, DejaVu fonts
Storage	Generated PNG panels and PDFs on disk	File system